# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 13: **Two Variables, One Line** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The line $4x + 3y = 9$ is parallel to the line $kx + 15y = 11$. The value of $k$ is: 1

**(a)** $20$ **(b)** $22$ **(c)** $21$ **(d)** $19$

**2.** The points $(-10, -3)$, $(-7, 9)$ and $(-4, k)$ are collinear. The value of $k$ is: 1

**(a)** $25$ **(b)** $21$ **(c)** $22$ **(d)** $20$

**3.** The line $kx + 2y = 12$ passes through $(1, 1)$. Its slope is: 1

**(a)** $-3$ **(b)** $-6$ **(c)** $-4$ **(d)** $-5$

**4.** The line $7x + 18y = 126$ has intercepts $p$ on the $x$-axis and $q$ on the $y$-axis. The value of $p + q$ is: 1

**(a)** $25$ **(b)** $26$ **(c)** $24$ **(d)** $151$

**5.** Using $F = \frac{9}{5}C + 32$, the temperature $85°$C in degrees Fahrenheit is: 1

**(a)** $185$ **(b)** $186$ **(c)** $194$ **(d)** $184$

**6.** The slope of the line through $(22, -12)$ and $(4, -5)$ is: 1

**(a)** $-\frac{7}{18}$ **(b)** $\frac{29}{18}$ **(c)** $-\frac{25}{18}$ **(d)** $\frac{11}{18}$

**7.** The line $22x + 23y = 506$ cuts the axes at two points. The area of the triangle it forms with the axes (square units) is: 1

**(a)** $255$ **(b)** $254$ **(c)** $253$ **(d)** $252$

**8.** A taxi fare is ₹279 for 3 km and ₹567 for 9 km (fare is a linear function of distance). The fare for 40 km (in ₹) is: 1

**(a)** $2190$ **(b)** $2103$ **(c)** $2007$ **(d)** $2055$

**9.** **Assertion (A):** The line $2x + 3y = 12$ cuts the x-axis at $(6, 0)$. 1

**Reason (R):** Every point on the x-axis has x-coordinate 0.

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

**10.** **Assertion (A):** The equation $x = 5$, considered in two variables, represents a single point on the Cartesian plane. 1

**Reason (R):** The graph of $x = 5$ is a line parallel to the y-axis.

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Find the area of the triangle formed by the line $15x + 19y = 285$ and the coordinate axes. 2

**12.** Find the equation of the line through $(3, 5)$ and $(-9, -19)$ in the form $y = mx + c$. 2

**13.** The line $kx + 5y = 10$ passes through the point $(2, -2)$. Find $k$, the slope and the $y$-intercept of the line. 2

**14.** Find the area of the triangle formed by the line $16x + 15y = 240$ and the coordinate axes. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** A line cuts the $x$-axis at $(12, 0)$ and the $y$-axis at $(0, 7)$. Find its equation, its slope and the area of the triangle it makes with the axes. 3

**16.** A water supplier charges a fixed amount plus a rate per kilolitre. For 4 kL the bill is ₹62 and for 15 kL it is ₹150. Find the equation of the bill, and the usage whose bill is ₹206. Also find the bill for no usage. 3

**17.** Find $k$ if the points $(-26, 30)$, $(-25, 32)$ and $(-24, k)$ lie on one straight line. 3

**OR**

A line cuts the $x$-axis at $(6, 0)$ and the $y$-axis at $(0, 13)$. Find its equation, its slope and the area of the triangle it makes with the axes. 3

**18.** A water supplier charges a fixed amount plus a rate per kilolitre. For 2 kL the bill is ₹64 and for 12 kL it is ₹194. Find the equation of the bill, and the usage whose bill is ₹285. Also find the bill for no usage. 3

**OR**

Find $k$ if the points $(-1, -19)$, $(0, -21)$ and $(1, k)$ lie on one straight line. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** A gardener lays a water pipe along the line $2x + y = 8$ on a grid of the garden (1 unit = 1 m). 5

(i) Write the equation in the form $y = mx + c$ and state the slope.

(ii) Find the y-intercept.

(iii) Find the x-intercept.

(iv) Check whether the point $(3, 2)$ lies on the pipe, and find the y-coordinate of the point of the pipe with $x = -1$.

**20.** A line passes through $(-3, 0)$ and $(1, 8)$. Find its equation, then find $k$ if the point $(k, 20)$ lies on it, and the $x$-intercept. 5

**OR**

A taxi fare is ₹178 for 2 km and ₹618 for 12 km. Assuming the fare is linear in the distance, find the fare per km, the fixed charge and the fare for 20 km. 5

## END OF QUESTION PAPER — Total: 40 Marks